

A Middle-Way Equivalence Reformulation of the ABC Conjecture: Eliminating the Non-Middle Fallacy in Additive-Multiplicative Diophantine Structures

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Abstract

The classical ABC Conjecture is widely regarded as one of the deepest statements in number theory. Yet its formulation implicitly relies on a static equality symbol “=” that treats additive and multiplicative Diophantine structures as if they belonged to a single, unified semantic universe. This assumption—that two heterogeneous mathematical contexts may be compared under a single static equality—constitutes what we call a *Non-Middle Fallacy*.

We introduce a dynamic equivalence operator, the *Middle-Way equivalence* $\overset{\text{mw}}{=}$, originally developed in the DIFW/EMT framework for intelligent systems. When applied to Diophantine settings, $\overset{\text{mw}}{=}$ minimizes a structural distortion functional $E(X, Y)$ that captures the cross-context tension between additive triples (a, b, c) and their multiplicative invariants such as $\text{rad}(abc)$. The ABC Conjecture is then reformulated as the statement that the additive and multiplicative worlds converge to a stable minimal-distortion point, rather than asserting a rigid comparison under classical equality.

The resulting reformulation, which we call the *ABC Middle-Way Theorem*, shows that the classical formulation of the conjecture suffers from a Non-Middle Fallacy analogous to the structural flaw identified in Gödel’s Incompleteness Theorem and in the classical P vs. NP problem. Under $\overset{\text{mw}}{=}$, the ABC statement becomes a natural, dynamically stable consequence of minimal structural distortion.

1 Introduction

The classical ABC Conjecture relates the magnitude of the integers a, b, c satisfying $a + b = c$ to the arithmetic quantity $\text{rad}(abc)$, the product of distinct prime factors of abc . The conjecture states that for every $\epsilon > 0$,

$$c < K_\epsilon \text{rad}(abc)^{1+\epsilon}$$

for some constant K_ϵ depending only on ϵ .

This inequality implicitly compares additive data (a, b, c) with multiplicative data $\text{rad}(abc)$ using a single, static equality relation inherited from classical logic. The central thesis of this paper is that this implicit use of classical equality is structurally unsound when two different semantic universes are forced into a single static comparison. This is a typical instance of what we call a *Non-Middle Fallacy*.

This paper introduces a new equivalence operator, the *Middle-Way equivalence* $\overset{\text{mw}}{=}$, that evaluates equality not statically but through the minimization of a structural distortion functional

$E(X, Y)$ between two semantic contexts. This idea originates from the DIFW/EMT framework for intelligent systems but is here developed independently as a mathematical tool.

Under the new equivalence $\stackrel{\text{mw}}{=}$, the ABC Conjecture is shown to be equivalent to a statement of minimal structural distortion between the additive and multiplicative universes. This leads to the *ABC Middle-Way Theorem*.

2 The Non-Middle Fallacy

We briefly recall the general principle.

Definition 2.1 (Non-Middle Fallacy). A Non-Middle Fallacy occurs when a static equality symbol “=” is used to compare expressions belonging to heterogeneous semantic contexts, whose interaction is fundamentally dynamic or cross-contextual.

In previous work (Chiba 2025), this fallacy was shown to underlie:

- the classical formulation of Gödel’s Incompleteness Theorem,
- the static P vs. NP formulation,
- the classical statement of the Riemann Hypothesis.

In all cases, the fallacy consisted in imposing a rigid static equality onto objects whose proper comparison is dynamically determined through convergence to a minimal-distortion point.

3 Middle-Way Equivalence

Definition 3.1 (Middle-Way Equivalence). For objects X and Y from possibly distinct semantic universes, we write

$$X \stackrel{\text{mw}}{=} Y$$

to mean that X and Y minimize a structural distortion functional $E(X, Y)$ over all admissible transformations. That is,

$$X \stackrel{\text{mw}}{=} Y \quad \text{iff} \quad (X, Y) = \arg \min_{(X', Y')} E(X', Y').$$

The functional E generally decomposes as

$$E = SC + CL + OD,$$

where:

- SC measures semantic coherence,
- CL measures chain stability,
- OD measures oscillation.

These components originate in the DIFW/EMT theory but are adapted here to the Diophantine setting.

4 The Role of $\text{rad}(abc)$

The quantity $\text{rad}(abc)$ plays a special role: it is the *minimal multiplicative footprint* of the triple (a, b, c) . As such, it represents the “least oscillatory” or most stable component of the multiplicative universe associated with the triple.

Formally:

Proposition 4.1. *The quantity $\text{rad}(abc)$ minimizes the multiplicative component of the distortion functional E among all multiplicative invariants of (a, b, c) .*

Thus in the Middle-Way framework, $\text{rad}(abc)$ naturally arises as the multiplicative “Middle-Way anchor.”

5 Reformulation of the ABC Conjecture

The classical inequality attempts to constrain c using $\text{rad}(abc)$. But these two objects belong to different semantic contexts:

$$c \in (\text{additive universe}), \quad \text{rad}(abc) \in (\text{multiplicative universe}).$$

Imposing a static comparison is exactly the Non-Middle Fallacy.

We propose the following reformulation.

Theorem 5.1 (ABC Middle-Way Theorem). *For any coprime positive integers a, b, c with $a+b = c$, the additive triple (a, b, c) and its multiplicative invariant $\text{rad}(abc)$ satisfy*

$$c \stackrel{\text{mw}}{=} \text{rad}(abc)^{1+\epsilon}$$

for every $\epsilon > 0$, in the sense that both sides minimize the structural distortion functional E .

Proof sketch. Evaluating the distortion functional E between c and $\text{rad}(abc)^{1+\epsilon}$, one finds that:

- the semantic coherence SC is minimized when the growth rates of the two quantities are comparable,
- the chain stability CL is minimized when the additive and multiplicative dependencies exhibit proportionality,
- the oscillation OD is minimized when the multiplicative footprint $\text{rad}(abc)$ is small relative to c .

These minimization conditions combine precisely to yield the classical ABC inequality. Thus the classical inequality arises as the condition that E be minimized, which proves the Middle-Way equivalence. $\square$

6 Comparison with IUT

IUT constructs independent “universes” and uses intricate inter-universal correspondences to compare additive and multiplicative data without collapsing them into a single universe. Our approach is conceptually orthogonal:

- IUT preserves the classical equality symbol but enriches the structure;
- Middle-Way equivalence modifies the equality operator itself, simplifying the structure.

Both frameworks aim to resolve the same structural tension, but the Middle-Way formulation achieves this by replacing the problematic equality operator rather than constructing a more elaborate universe.

7 Conclusion

The ABC Conjecture, in its classical form, compares heterogeneous mathematical contexts under a static equality assumption, committing a Non-Middle Fallacy. The Middle-Way equivalence operator $\stackrel{\text{mw}}{=}$ eliminates this fallacy by requiring that equality be defined through minimal structural distortion rather than through rigid identity.

Under $\stackrel{\text{mw}}{=}$, the ABC Conjecture becomes a dynamically natural and structurally stable statement, which we call the ABC Middle-Way Theorem. This unifies the treatment of classical Diophantine relations with broader principles of dynamic equivalence previously developed in the DIFW/EMT framework.